

OAKLAND TROY, MI, USA

WMO: 720113

Lat: 42.543N

Lon: 83.178W

Elev: 716

StdP: 14.32

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 54829

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.5	9.3	-6.5	4.0	7.4	-1.6	5.2	11.2	23.5	27.9	20.5	28.5	6.6	210	0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	91.2	70.9	89.1	70.0	86.3	68.9	74.9	84.6	73.4	83.3	72.0	81.3	9.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.2	122.4	79.5	70.4	115.0	78.2	69.0	109.6	77.3	38.8	84.6	37.4	82.9	36.1	81.7	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.0	17.7	15.8	DB	-1.1	95.5	6.5	2.3	-5.8	97.2	-9.6	98.5	-13.2	99.8	-17.9	101.5	(4)
			WB	-2.1	77.7	6.2	1.4	-6.6	78.7	-10.2	79.5	-13.7	80.3	-18.2	81.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.5	26.9	28.0	37.5	49.1	61.3	70.5	75.0	73.4	66.3	54.1	41.7	32.4
	DBStd	19.09	10.42	10.00	10.57	8.78	8.85	6.43	5.53	5.21	7.31	8.48	8.81	8.42
	HDD50	2750	718	617	412	120	10	0	0	0	0	50	276	547
	HDD65	5890	1182	1035	853	480	180	21	2	5	71	352	699	1010
	CDD50	3293	1	1	25	93	361	615	775	726	490	178	26	2
	CDD65	959	0	0	1	3	66	186	312	266	110	15	0	0
	CDH74	8354	0	0	16	60	662	1628	2906	2146	819	115	2	0
	CDH80	2793	0	0	3	7	204	563	1070	673	253	20	0	0
Wind	WSAvg	6.8	7.8	7.7	7.5	8.1	6.6	6.2	5.6	5.3	5.4	6.6	7.3	7.4
Precipitation	PrecAvg	33.3	2.1	1.8	2.2	3.0	3.8	3.3	3.5	3.1	3.1	2.7	2.5	2.2
	PrecMax	43.3	3.8	3.5	3.9	6.2	8.7	5.2	6.7	5.6	7.0	7.3	4.4	3.8
	PrecMin	23.5	0.6	0.6	0.8	0.6	1.7	1.3	0.7	0.7	0.3	0.7	0.7	0.9
	PrecStd	4.7	0.9	0.9	1.0	1.2	1.6	1.2	1.5	1.1	1.7	1.4	1.1	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.0	60.8	75.0	81.1	89.9	93.4	95.5	92.8	91.3	82.6	70.7	58.7
		MCWB	54.3	53.1	61.2	60.6	68.4	70.8	73.3	71.7	72.4	69.2	56.7	55.3
	2%	DB	51.1	52.3	64.5	74.3	85.5	90.4	91.2	89.6	87.1	77.2	63.4	53.2
		MCWB	47.5	48.1	54.0	56.9	65.9	69.9	71.3	70.0	70.4	64.4	53.7	50.0
	5%	DB	45.3	45.6	59.3	70.0	81.6	86.4	89.2	86.5	82.1	72.3	59.1	47.9
		MCWB	41.8	40.6	51.1	54.6	64.6	68.5	70.4	69.3	67.3	61.7	51.3	44.2
	10%	DB	40.3	40.8	54.1	64.2	77.0	82.4	86.1	83.8	78.7	68.2	54.9	43.7
		MCWB	37.3	36.3	46.6	51.9	62.3	66.1	69.2	68.1	65.9	59.4	48.4	39.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.1	54.9	61.9	63.7	71.9	75.0	77.5	76.5	75.1	70.8	60.2	56.1
		MCDB	56.7	57.6	75.4	75.2	82.6	86.7	88.7	86.8	85.8	79.2	65.3	58.4
	2%	WB	47.8	47.8	57.1	60.2	69.6	73.0	75.3	74.3	73.0	67.6	55.7	50.3
		MCDB	49.4	51.6	63.2	69.3	80.3	84.4	84.8	83.1	82.3	75.0	61.8	52.4
	5%	WB	41.8	40.5	52.5	57.0	67.5	71.4	73.6	72.4	70.9	63.8	52.3	44.9
		MCDB	45.1	44.7	57.8	66.9	77.9	81.8	83.6	81.1	78.5	69.8	58.1	47.4
	10%	WB	37.6	36.7	46.4	53.6	64.8	69.4	72.1	71.0	68.6	60.2	49.2	40.2
		MCDB	40.1	40.5	52.8	63.3	74.6	78.5	81.3	79.6	75.6	66.1	54.9	43.5
Mean Daily Temperature Range	5% DB	MDBR	11.8	12.8	15.2	18.5	19.3	18.4	18.0	16.9	17.2	15.7	13.3	9.7
		MCDBR	16.4	18.6	23.1	26.5	23.5	21.6	20.3	19.4	21.5	21.3	20.4	15.3
		MCWBR	14.5	15.3	15.5	14.5	10.6	9.0	7.6	7.1	8.3	11.3	13.8	13.6
	5% WB	MCDBR	16.3	18.2	22.2	23.8	20.1	18.9	17.5	16.2	18.0	18.9	17.9	14.7
		MCWBR	15.4	16.0	16.3	16.0	10.6	10.0	7.9	7.4	8.7	12.0	14.1	14.3
Clear-Sky Solar Irradiance	taub	0.299	0.333	0.349	0.394	0.423	0.443	0.433	0.426	0.387	0.358	0.319	0.306	
	taud	2.395	2.280	2.362	2.254	2.223	2.203	2.253	2.263	2.354	2.419	2.493	2.418	
	Ebn at Noon	267	272	283	277	271	265	266	264	267	262	259	253	
	Edh at Noon	25	33	35	42	44	45	43	41	35	29	23	23	
All-Sky Solar Radiation	RadAvg	472	750	1118	1464	1708	1930	1932	1670	1347	874	540	383	
	RadStd	47	87	115	129	148	119	94	66	120	76	69	30	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (60 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page